

Weak Formulations of Scalar Elliptic Problems: Boundedness, Coercivity, and How the Function Space Is Chosen

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Abstract

These notes work through three scalar elliptic boundary value problems. For each one I derive the weak formulation and then prove that the resulting bilinear form is bounded and coercive. Two things are worth watching. The first is that the function space is not chosen in advance — it falls out of the weak formulation once we ask what is needed to make the integrals finite. The second is that boundedness and coercivity point in opposite directions and need different tools: boundedness is an upper bound on $a(u, v)$ and comes from Cauchy–Schwarz, while coercivity is a lower bound on $a(u, u)$ and comes from setting $v = u$ and then finding the weakest coefficient. The third problem is the microrotation equation of the micropolar system with the coupling switched off, so the last result here is used directly later on.

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1 Notation and assumptions

Throughout, $\Omega \subset \mathbb{R}^n$ is a bounded Lipschitz domain with boundary $\partial\Omega$. All derivatives are weak derivatives. I write

$$\|u\|_{L^2} = \left(\int_{\Omega} |u|^2 \, dx \right)^{1/2}, \quad \|u\|_{H^1}^2 = \|u\|_{L^2}^2 + \|\nabla u\|_{L^2}^2,$$

and $H_0^1(\Omega)$ for the closure of $C_c^\infty(\Omega)$ in the $H^1(\Omega)$ norm. When $n = 1$ and $\Omega = (0, 1)$ I write u' instead of ∇u .

Two results are used over and over.

Lemma 1.1 (Cauchy–Schwarz). *For $u, v \in L^2(\Omega)$,*

$$\left| \int_{\Omega} uv \, dx \right| \leq \|u\|_{L^2} \|v\|_{L^2}.$$

Lemma 1.2 (Poincaré–Friedrichs). *There is a constant $C_\Omega > 0$, depending only on Ω , with*

$$\|u\|_{L^2} \leq C_\Omega \|\nabla u\|_{L^2} \quad \text{for every } u \in H_0^1(\Omega).$$

Remark 1.3. Lemma 1.2 is false on $H^1(\Omega)$. Take $u \equiv 1$. Then $\nabla u = 0$, so the right-hand side is zero, but $\|u\|_{L^2} = |\Omega|^{1/2} \neq 0$. What makes the inequality work on $H_0^1(\Omega)$ is that the boundary condition rules out non-zero constants. So the boundary condition is not a side detail. It is the hypothesis the whole existence theory rests on, and I come back to this point at the end of Section 4.

Remark 1.4. One consequence of Lemma 1.2 is that $\|\nabla \cdot\|_{L^2}$ is itself a norm on $H_0^1(\Omega)$, equivalent to $\|\cdot\|_{H^1}$. Many papers use it without saying so, which is confusing the first time you meet it.

2 The one-dimensional model problem

2.1 From the strong form to the weak form

Take the two-point boundary value problem

$$-u'' = f \quad \text{on } (0, 1), \quad u(0) = u(1) = 0. \tag{1}$$

Let v be a test function with $v(0) = v(1) = 0$. Multiply (1) by v and integrate over $(0, 1)$:

$$-\int_0^1 u'' v \, dx = \int_0^1 f v \, dx.$$

Integrate the left-hand side by parts once:

$$-\int_0^1 u'' v \, dx = -\left[u' v \right]_0^1 + \int_0^1 u' v' \, dx = \int_0^1 u' v' \, dx,$$

where the boundary term drops out because $v(0) = v(1) = 0$. It is worth being careful about which function is doing that work: it is v , not u' . There is no reason for u' to vanish at the endpoints, and in general it does not.

So the weak problem is: find $u \in H_0^1(0, 1)$ with

$$a(u, v) := \int_0^1 u' v' \, dx = \int_0^1 f v \, dx =: F(v) \quad \text{for all } v \in H_0^1(0, 1). \tag{2}$$

The trade we made is worth naming. The classical form needs two derivatives of u and none of v ; the weak form needs one of each. We gave away regularity of v , which costs nothing because we choose v ourselves and can always choose it smooth, in order to buy back regularity of u , which is what we actually care about.

2.2 Why the space is H_0^1

The classical formulation (1) asks for $u \in C^2$. That demand comes from the way we wrote the equation down, not from the problem being modelled, and it throws away solutions that exist in reality. If f has a jump — a bar under a point load, or a change of material halfway along — then no C^2 function satisfies (1). The bar still bends. So a theory built on C^2 would tell us there is no solution to a problem whose solution we can see.

The weak form (2) only involves u' , so the right space is found by asking what is needed for its integrals to make sense. By Lemma 1.1, $\int_0^1 u'v'$ is finite as soon as $u', v' \in L^2(0, 1)$, and we needed $v(0) = v(1) = 0$ to kill the boundary term. That gives

$$H_0^1(0, 1) = \{ u \in L^2(0, 1) : u' \in L^2(0, 1), u(0) = u(1) = 0 \}.$$

Note the direction of that argument. The space was built out of the requirement; it was not picked first and then imposed.

The choice is squeezed from both sides. Go bigger — drop the condition $u' \in L^2$ and work in $L^2(0, 1)$ alone — and $\int_0^1 u'v'$ may be infinite, so the equation stops meaning anything. Go smaller — C^2 , or $H^2 \cap H_0^1$ — and we are assuming more regularity than the formulation ever asked for, which throws out the point-load solution above. H_0^1 is the largest space on which (2) still makes sense, and that is what makes it the right one.

Remark 2.1. In one dimension $H^1(0, 1) \hookrightarrow C[0, 1]$, so writing $u(0) = u(1) = 0$ is fine. In two dimensions this needs more care, because $\partial\Omega$ has measure zero and $H^1(\Omega)$ functions are only defined up to null sets. See Remark 3.1.

2.3 Boundedness

Proposition 2.2. *For all $u, v \in H_0^1(0, 1)$ we have $|a(u, v)| \leq \|u\|_{H^1} \|v\|_{H^1}$.*

Proof. Cauchy–Schwarz (Lemma 1.1) applied to u' and v' gives

$$|a(u, v)| = \left| \int_0^1 u'v' \, dx \right| \leq \|u'\|_{L^2} \|v'\|_{L^2}. \quad (3)$$

Now $\|u\|_{H^1}^2 = \|u\|_{L^2}^2 + \|u'\|_{L^2}^2$ and $\|u\|_{L^2}^2 \geq 0$, so $\|u'\|_{L^2}^2 \leq \|u\|_{H^1}^2$, i.e.

$$\|u'\|_{L^2} \leq \|u\|_{H^1}, \quad \text{and in the same way} \quad \|v'\|_{L^2} \leq \|v\|_{H^1}. \quad (4)$$

Putting (3) and (4) together finishes it, with constant $C = 1$. $\square$

Remark 2.3. Poincaré does not appear here, and it could not have. Boundedness needs the seminorm controlled by the full norm, $\|u'\|_{L^2} \leq \|u\|_{H^1}$, which is free. Poincaré gives control in the opposite direction, $\|u\|_{H^1} \lesssim \|u'\|_{L^2}$, which is not free and is what coercivity needs. Two opposite directions, two different jobs. Mixing them up is the easiest mistake to make here, and I made it the first time.

2.4 Coercivity

Proposition 2.4. *For all $u \in H_0^1(0, 1)$,*

$$a(u, u) \geq \alpha \|u\|_{H^1}^2, \quad \alpha = \frac{1}{C_\Omega^2 + 1}.$$

Proof. Put $v = u$. This gives an identity, not an estimate:

$$a(u, u) = \int_0^1 (u')^2 \, dx = \|u'\|_{L^2}^2. \quad (5)$$

There is nothing to bound at this step — $\int_0^1 (u')^2$ is $\|u'\|_{L^2}^2$, straight from the definition of the norm. This is why setting $v = u$ is the whole trick: Cauchy–Schwarz exists to handle two different functions, and with the same function twice there is nothing left to estimate.

By Lemma 1.2, $\|u\|_{L^2}^2 \leq C_\Omega^2 \|u'\|_{L^2}^2$, so

$$\|u\|_{H^1}^2 = \|u\|_{L^2}^2 + \|u'\|_{L^2}^2 \leq C_\Omega^2 \|u'\|_{L^2}^2 + \|u'\|_{L^2}^2 = (C_\Omega^2 + 1) \|u'\|_{L^2}^2. \quad (6)$$

Rearranging (6),

$$\|u'\|_{L^2}^2 \geq \frac{1}{C_\Omega^2 + 1} \|u\|_{H^1}^2, \quad (7)$$

and combining this with (5) gives the result. $\square$

Remark 2.5 (Sanity check). The constant is $\alpha = (C_\Omega^2 + 1)^{-1}$, so α gets smaller as C_Ω gets bigger, which means coercivity gets weaker on domains with a large Poincaré constant. Since C_Ω grows with the size of Ω , this is saying something I find intuitive: on a long interval the two endpoint conditions do not constrain the function much, because a function with a small derivative still has room to wander a long way from zero in the middle. On a short interval the same two conditions clamp it. So α is measuring how hard the boundary condition is working, which lines up with Remark 1.3.

2.5 Stability of the solution

Proposition 2.6. *Let $f \in L^2(0, 1)$ and let $u \in H_0^1(0, 1)$ solve (2). Then*

$$\|u\|_{H^1} \leq \frac{1}{\alpha} \|f\|_{L^2}, \quad \alpha = \frac{1}{C_\Omega^2 + 1}.$$

Proof. Take $v = u$ in (2), so $a(u, u) = \int_0^1 f u \, dx$. Bound the left side from below with Proposition 2.4 and the right side from above with Lemma 1.1:

$$\alpha \|u\|_{H^1}^2 \leq a(u, u) = \int_0^1 f u \, dx \leq \|f\|_{L^2} \|u\|_{L^2} \leq \|f\|_{L^2} \|u\|_{H^1}.$$

If $u = 0$ there is nothing to prove. Otherwise divide by $\|u\|_{H^1} > 0$. $\square$

Remark 2.7. This says: bounded data gives a bounded solution, with a constant we can write down. That is stability, and it is the continuous version of what we want from a numerical scheme — that it should not amplify perturbations in the data. Worth noticing that it came from coercivity and not from boundedness. Coercivity is the property that does the real work; boundedness is a technical hypothesis that is almost always easy.

3 The Dirichlet problem in two dimensions

3.1 From the strong form to the weak form

Now let $\Omega \subset \mathbb{R}^2$ and take

$$-\Delta u = f \text{ in } \Omega, \quad u = 0 \text{ on } \partial\Omega. \quad (8)$$

The tool is Green’s first identity: for $u \in H^2(\Omega)$ and $v \in H^1(\Omega)$,

$$\int_\Omega (\Delta u) v \, dx = \int_{\partial\Omega} \frac{\partial u}{\partial n} v \, ds - \int_\Omega \nabla u \cdot \nabla v \, dx, \quad (9)$$

with $\frac{\partial u}{\partial n} = \nabla u \cdot \mathbf{n}$ and $\mathbf{n}$ the outward unit normal. This is integration by parts in n dimensions, and it lines up term by term with the one-dimensional case: $\int_\Omega (\Delta u) v$ against $\int_0^1 u'' v$, the

boundary integral against the evaluation $[u'v]_0^1$, and $\int_{\Omega} \nabla u \cdot \nabla v$ against $\int_0^1 u'v'$. In 1D the “boundary” is two points and the outward normal derivative is $+u'(1)$ and $-u'(0)$, which is exactly what the square bracket computes.

Multiply (8) by $v \in H_0^1(\Omega)$, integrate over Ω , and apply (9):

$$-\int_{\Omega} (\Delta u)v \, dx = -\int_{\partial\Omega} \frac{\partial u}{\partial n} v \, ds + \int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} \nabla u \cdot \nabla v \, dx,$$

the boundary integral vanishing because v has zero trace. Again it is v that does this, not $\frac{\partial u}{\partial n}$.

The weak problem: find $u \in H_0^1(\Omega)$ with

$$a(u, v) := \int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx \quad \text{for all } v \in H_0^1(\Omega), \quad (10)$$

where $\nabla u \cdot \nabla v = \partial_x u \partial_x v + \partial_y u \partial_y v$. In one dimension that sum has a single term and reads $u'v'$; in two dimensions it has two. That is the only change.

Remark 3.1. “ $v = 0$ on $\partial\Omega$ ” cannot be read pointwise here. The boundary has measure zero in $\mathbb{R}^2$, and elements of $H^1(\Omega)$ are equivalence classes modulo null sets, so the value of v at any particular point is not well defined. The statement is made precise by the trace theorem: the restriction map $v \mapsto v|_{\partial\Omega}$, defined on $C(\overline{\Omega}) \cap H^1(\Omega)$, extends to a bounded operator on $H^1(\Omega)$, and $H_0^1(\Omega)$ is exactly the kernel of that operator. I am citing this, not proving it.

Remark 3.2. Identity (9) needs $u \in H^2(\Omega)$ so that Δu is a function. That regularity is used only to get to (10) from (8). The weak problem itself only needs $u \in H_0^1(\Omega)$, and the analysis below is done there. Deriving under classical regularity and then dropping it is the standard order of business, and it is easy to be confused by it the first time.

3.2 Boundedness and coercivity

Proposition 3.3. *For all $u, v \in H_0^1(\Omega)$, $|a(u, v)| \leq \|u\|_{H^1} \|v\|_{H^1}$.*

Proof. Cauchy–Schwarz is used twice: once pointwise on the vectors ∇u and ∇v , then once on the integral via Lemma 1.1. That gives

$$|a(u, v)| = \left| \int_{\Omega} \nabla u \cdot \nabla v \, dx \right| \leq \int_{\Omega} |\nabla u| |\nabla v| \, dx \leq \|\nabla u\|_{L^2} \|\nabla v\|_{L^2}.$$

Then $\|\nabla u\|_{L^2} \leq \|u\|_{H^1}$ and $\|\nabla v\|_{L^2} \leq \|v\|_{H^1}$, exactly as in (4). $\square$

Proposition 3.4. *For all $u \in H_0^1(\Omega)$, $a(u, u) \geq \alpha \|u\|_{H^1}^2$ with $\alpha = (C_{\Omega}^2 + 1)^{-1}$.*

Proof. Setting $v = u$ gives $a(u, u) = \int_{\Omega} |\nabla u|^2 \, dx = \|\nabla u\|_{L^2}^2$. Poincaré holds on $H_0^1(\Omega)$ in any dimension, so

$$\|u\|_{H^1}^2 = \|u\|_{L^2}^2 + \|\nabla u\|_{L^2}^2 \leq (C_{\Omega}^2 + 1) \|\nabla u\|_{L^2}^2,$$

and the rest is as in Proposition 2.4. $\square$

Remark 3.5. These two proofs are the one-dimensional ones with u' replaced by ∇u , and the constants come out the same. The dimension could have caused trouble in two places — the integration by parts, and the meaning of the boundary condition — and both were dealt with by quoting a theorem (Green’s identity, and the trace theorem). Neither changed the shape of the argument. The point of this section is that nothing new happens, which is itself worth knowing: the machinery is dimension-independent.

4 The reaction–diffusion problem

4.1 From the strong form to the weak form

Let $c, \kappa > 0$ and take

$$-c \Delta u + \kappa u = f \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega. \quad (11)$$

Multiply by $v \in H_0^1(\Omega)$ and integrate:

$$-c \int_{\Omega} (\Delta u) v \, dx + \kappa \int_{\Omega} u v \, dx = \int_{\Omega} f v \, dx.$$

Apply (9) to the first term; its boundary contribution vanishes for the same reason as before. The second term is left alone — there is no derivative in it, so there is nothing to integrate by parts. This gives: find $u \in H_0^1(\Omega)$ with

$$a(u, v) := c \int_{\Omega} \nabla u \cdot \nabla v \, dx + \kappa \int_{\Omega} u v \, dx = \int_{\Omega} f v \, dx \quad \text{for all } v \in H_0^1(\Omega). \quad (12)$$

4.2 Coercivity, this time without Poincaré

Proposition 4.1. *For all $u \in H_0^1(\Omega)$,*

$$a(u, u) \geq \min(c, \kappa) \|u\|_{H^1}^2.$$

Proof. Set $v = u$ in (12). As before this is an identity:

$$a(u, u) = c \int_{\Omega} |\nabla u|^2 \, dx + \kappa \int_{\Omega} u^2 \, dx = c \|\nabla u\|_{L^2}^2 + \kappa \|u\|_{L^2}^2.$$

Both pieces of $\|u\|_{H^1}^2$ are already sitting there. The only difference is the coefficient in front of each. Replacing both coefficients by the smaller of the two,

$$a(u, u) \geq \min(c, \kappa) \left(\|\nabla u\|_{L^2}^2 + \|u\|_{L^2}^2 \right) = \min(c, \kappa) \|u\|_{H^1}^2. \quad \square$$

Remark 4.2. Poincaré is never used. The zeroth-order term hands us $\|u\|_{L^2}^2$ for free, and that is precisely the piece Poincaré had to manufacture in Propositions 2.4 and 3.4.

Remark 4.3 (Sanity check). As $\kappa \rightarrow 0$ the constant $\min(c, \kappa)$ goes to zero and this proof collapses. That is not a flaw in the proof — it is the proof telling us which regime it belongs to. In that limit (11) turns back into (8), which is still coercive, but only via Poincaré and therefore only via the boundary condition. So the two proofs are identifying two different mechanisms for controlling $\|u\|_{L^2}^2$. When $\kappa > 0$ the reaction term controls it directly and the boundary condition is not needed for that job. When $\kappa = 0$ the boundary condition is all there is, and it acts through C_{Ω} . Something has to control $\|u\|_{L^2}^2$; the question is only what.

5 Summary

Problem	Space	Coercivity constant	What supplies coercivity
$-u'' = f$	$H_0^1(0, 1)$	$(C_{\Omega}^2 + 1)^{-1}$	Poincaré
$-\Delta u = f$	$H_0^1(\Omega)$	$(C_{\Omega}^2 + 1)^{-1}$	Poincaré
$-c\Delta u + \kappa u = f$	$H_0^1(\Omega)$	$\min(c, \kappa)$	the reaction term

Two things to take away.

First, in every case the space came out of the weak formulation rather than being chosen beforehand. The recipe was always the same: write the weak form, look at the integrals in it, and ask what the weakest condition on u is that keeps them finite. Section 2.2 then argued that this space is optimal — anything larger and the formulation stops making sense, anything smaller and we lose solutions that genuinely exist.

Second, boundedness and coercivity are opposite kinds of statement and need different tools. Boundedness bounds $a(u, v)$ from above, with two independent arguments, and follows from Cauchy–Schwarz plus the free estimate $\|\nabla u\|_{L^2} \leq \|u\|_{H^1}$. Coercivity bounds $a(u, u)$ from below, with the same argument twice; setting $v = u$ produces an identity, and after that the only question is what bounds $\|u\|_{H^1}^2$ in terms of what the identity gave us. Whether Poincaré is needed for that depends entirely on whether the equation has a zeroth-order term. Once these two are in hand, Lax–Milgram gives existence and uniqueness, which is the point of the whole exercise.

Where this connects to the micropolar problem

Problem (11) is the microrotation equation of the two-dimensional stationary linear micropolar system,

$$-(c_a + c_d) \Delta w + 4\nu_r w = 2\nu_r \operatorname{curl} \mathbf{u} + g,$$

with the coupling term $2\nu_r \operatorname{curl} \mathbf{u}$ switched off, so that $c = c_a + c_d$ and $\kappa = 4\nu_r$. Proposition 4.1 therefore identifies the term that will have to absorb the indefinite coupling contribution when the full system is analysed.

It also shows that the vortex viscosity ν_r enters the problem twice, and in competing ways: it supplies the zeroth-order term that helps coercivity, and it multiplies the coupling term that hurts it. Sorting out that competition — which requires Young’s inequality with a free parameter, chosen so that the coupling is absorbed without eating the terms that were controlling things — is the subject of the next set of notes.